

Shiv Nadar University Chennai

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3801 – Web Technologies Lab	AY: 2026–27

Assignment - 4 Online Shopping Product Catalog

1 Aim

To develop a React-based Online Shopping Product Catalog that demonstrates dynamic searching, filtering, and conditional rendering using React state management. Features Search products by name. Filter by category (Electronics, Clothing, Grocery). Filter by price range. Show "Out of Stock" if quantity is 0. Display product count. Concepts useState Multiple filters filter() map() Conditional rendering

2 Code Implementation

2.1 main.jsx

```
App.jsx  main.jsx  # App.css
src > main.jsx
1  import { StrictMode } from "react";
2  import { createRoot } from "react-dom/client";
3  import App from "./App.jsx";
4
5  createRoot(document.getElementById("root")).render(
6    <StrictMode>
7      <App />
8    </StrictMode>
9  );
```

2.2 App.jsx

```
App.jsx  main.jsx  App.css
src > App.jsx > default
1  import { useState } from "react";
2  import "./App.css";
3
4  const products = [
5    {
6      id: 1,
7      name: "Laptop",
8      category: "Electronics",
9      price: 55000,
10     quantity: 5,
11     image: "/images/laptop-floating-angled-open.webp"
12   },
13   {
14     id: 2,
15     name: "Smartphone",
16     category: "Electronics",
17     price: 25000,
18     quantity: 8,
19     image: "/images/SmartPhone.webp"
20   },
21   {
22     id: 3,
23     name: "Headphones",
24     category: "Electronics",
25     price: 2800,
26     quantity: 0,
27     image: "/images/headphones.webp"
28   },
29   {
30     id: 4,
31     name: "T-Shirt",
32     category: "Clothing",
33     price: 799,
34     quantity: 15,
35     image: "/images/t-shirts.webp"
36   },
37   {
38     id: 5,
39     name: "Jeans",
40     category: "Clothing",
41     price: 1499,
42     quantity: 6,
43     image: "/images/jeans.webp"
44   },
45   {
46     id: 6,
47     name: "Jacket",
48     category: "Clothing",
49     price: 2499,
50     quantity: 0,
51     image: "/images/jacket.webp"
52   },
53   {
54     id: 7,
55     name: "Rice",
56     category: "Grocery",
57     price: 600,
58     quantity: 20,
59     image: "/images/Rice.webp"
60   },
61   {
62     id: 8,
63     name: "Milk",
64     category: "Grocery",
65     price: 60,
66     quantity: 30,
67     image: "/images/Milk.webp"
68   },
69   {
70     id: 9,
71     name: "Cooking Oil",
72     category: "Grocery",
73     price: 180,
74     quantity: 0,
75     image: "/images/Oil.webp"
76   }
77 ];
```

```

src > App.jsx > default
79 function App() {
81   // =====
82   // STATE
83   // =====
84
85   const [search, setSearch] = useState("");
86   const [category, setCategory] = useState("All");
87   const [minPrice, setMinPrice] = useState(0);
88   const [maxPrice, setMaxPrice] = useState(100000);
89
90   // Theme state
91   const [darkMode, setDarkMode] = useState(false);
92
93
94   // =====
95   // FILTER PRODUCTS
96   // =====
97
98   const filteredProducts = products.filter(product => {
99
100     const matchesSearch =
101       product.name
102         .toLowerCase()
103         .includes(search.toLowerCase());
104
105     const matchesCategory =
106       category === "All" ||
107       product.category === category;
108
109     const matchesPrice =
110       product.price >= minPrice &&
111       product.price <= maxPrice;
112
113     return (
114       matchesSearch &&
115       matchesCategory &&
116       matchesPrice
117     );
118   });
119
120
121   return (
122
123     <div className={darkMode ? "app dark" : "app"}>
124
125       /* =====
126       | HEADER
127       | ===== */
128
129       <header>
130
131         <h1>
132           🛒 Online Shopping Catalog
133         </h1>
134
135         <p>
136           Search and filter products
137         </p>
138
139         /* THEME BUTTON */
140
141         <button
142           className="theme-button"
143           onClick={() => setDarkMode(!darkMode)}
144         >
145           {darkMode
146             ? "☀️ Light Mode"
147             : "🌙 Dark Mode"}
148         </button>
149
150       </header>
151

```

```

App.jsx main.jsx # App.css
src > App.jsx > default
79 function App() {
153
154     FILTER SECTION
155     ----- */
156
157     <div className="filters">
158
159         /* SEARCH */
160
161         <div className="filter-group">
162
163             <label>
164                 Search Product
165             </label>
166
167             <input
168                 type="text"
169                 placeholder="Search by name..."
170                 value={search}
171                 onChange={(e) =>
172                     setSearch(e.target.value)
173                 }
174             />
175
176         </div>
177
178         /* CATEGORY */
179
180         <div className="filter-group">
181
182             <label>
183                 Category
184             </label>
185
186             <select
187                 value={category}
188                 onChange={(e) =>
189                     setCategory(e.target.value)
190                 }
191             >
192
193                 <option value="All">
194                     All Categories
195                 </option>
196
197                 <option value="Electronics">
198                     Electronics
199                 </option>
200
201                 <option value="Clothing">
202                     Clothing
203                 </option>
204
205                 <option value="Grocery">
206                     Grocery
207                 </option>
208
209             </select>
210
211         </div>
212
213         /* MINIMUM PRICE */
214
215         <div className="filter-group">
216
217             <label>
218                 Minimum Price
219             </label>
220
221             <input
222                 type="number"
223                 value={minPrice}
224                 onChange={(e) =>
225                     setMinPrice(
226                         Number(e.target.value)
227                     )
228                 }
229             />
230
231         </div>
232
233         /* MAXIMUM PRICE */
234
235         <div className="filter-group">
236
237             <label>
238                 Maximum Price
239             </label>
240
241             <input
242                 type="number"
243                 value={maxPrice}
244                 onChange={(e) =>
245                     setMaxPrice(
246                         Number(e.target.value)
247                     )
248                 }
249             />
250
251         </div>
252     </div>
253
254     </div>
255
256     </div>
257
258     </div>
259
260     </div>
261
262     </div>
263
264     </div>
265
266     </div>
267
268     </div>
269
270     </div>
271
272     </div>
273
274     </div>
275
276     </div>
277
278     </div>
279
280     </div>
281
282     </div>
283
284     </div>
285
286     </div>
287
288     </div>
289
290     </div>
291
292     </div>
293
294     </div>
295
296     </div>
297
298     </div>
299
300     </div>
301
302     </div>
303
304     </div>
305
306     </div>
307
308     </div>
309
310     </div>
311
312     </div>
313
314     </div>
315
316     </div>
317
318     </div>
319
320     </div>
321
322     </div>
323
324     </div>
325
326     </div>
327
328     </div>
329
330     </div>
331
332     </div>
333
334     </div>
335
336     </div>
337
338     </div>
339
340     </div>
341
342     </div>
343
344     </div>
345
346     </div>
347
348     </div>
349
350     </div>
351
352     </div>
353
354     </div>
355
356     </div>
357
358     </div>
359
360     </div>
361
362     </div>
363
364     </div>
365
366     </div>
367
368     </div>
369
370     </div>
371
372     </div>
373
374     </div>
375
376     </div>
377
378     </div>
379
380     </div>
381
382     </div>
383
384     </div>
385
386     </div>
387
388     </div>
389
390     </div>
391
392     </div>
393
394     </div>
395
396     </div>
397
398     </div>
399
400     </div>
401
402     </div>
403
404     </div>
405
406     </div>
407
408     </div>
409
410     </div>
411
412     </div>
413
414     </div>
415
416     </div>
417
418     </div>
419
420     </div>
421
422     </div>
423
424     </div>
425
426     </div>
427
428     </div>
429
430     </div>
431
432     </div>
433
434     </div>
435
436     </div>
437
438     </div>
439
440     </div>
441
442     </div>
443
444     </div>
445
446     </div>
447
448     </div>
449
450     </div>
451
452     </div>
453
454     </div>
455
456     </div>
457
458     </div>
459
460     </div>
461
462     </div>
463
464     </div>
465
466     </div>
467
468     </div>
469
470     </div>
471
472     </div>
473
474     </div>
475
476     </div>
477
478     </div>
479
480     </div>
481
482     </div>
483
484     </div>
485
486     </div>
487
488     </div>
489
490     </div>
491
492     </div>
493
494     </div>
495
496     </div>
497
498     </div>
499
500     </div>
501
502     </div>
503
504     </div>
505
506     </div>
507
508     </div>
509
510     </div>
511
512     </div>
513
514     </div>
515
516     </div>
517
518     </div>
519
520     </div>
521
522     </div>
523
524     </div>
525
526     </div>
527
528     </div>
529
530     </div>
531
532     </div>
533
534     </div>
535
536     </div>
537
538     </div>
539
540     </div>
541
542     </div>
543
544     </div>
545
546     </div>
547
548     </div>
549
550     </div>
551
552     </div>
553
554     </div>
555
556     </div>
557
558     </div>
559
560     </div>
561
562     </div>
563
564     </div>
565
566     </div>
567
568     </div>
569
570     </div>
571
572     </div>
573
574     </div>
575
576     </div>
577
578     </div>
579
580     </div>
581
582     </div>
583
584     </div>
585
586     </div>
587
588     </div>
589
590     </div>
591
592     </div>
593
594     </div>
595
596     </div>
597
598     </div>
599
600     </div>
601
602     </div>
603
604     </div>
605
606     </div>
607
608     </div>
609
610     </div>
611
612     </div>
613
614     </div>
615
616     </div>
617
618     </div>
619
620     </div>
621
622     </div>
623
624     </div>
625
626     </div>
627
628     </div>
629
630     </div>
631
632     </div>
633
634     </div>
635
636     </div>
637
638     </div>
639
640     </div>
641
642     </div>
643
644     </div>
645
646     </div>
647
648     </div>
649
650     </div>
651
652     </div>
653
654     </div>
655
656     </div>
657
658     </div>
659
660     </div>
661
662     </div>
663
664     </div>
665
666     </div>
667
668     </div>
669
670     </div>
671
672     </div>
673
674     </div>
675
676     </div>
677
678     </div>
679
680     </div>
681
682     </div>
683
684     </div>
685
686     </div>
687
688     </div>
689
690     </div>
691
692     </div>
693
694     </div>
695
696     </div>
697
698     </div>
699
700     </div>
701
702     </div>
703
704     </div>
705
706     </div>
707
708     </div>
709
710     </div>
711
712     </div>
713
714     </div>
715
716     </div>
717
718     </div>
719
720     </div>
721
722     </div>
723
724     </div>
725
726     </div>
727
728     </div>
729
730     </div>
731
732     </div>
733
734     </div>
735
736     </div>
737
738     </div>
739
740     </div>
741
742     </div>
743
744     </div>
745
746     </div>
747
748     </div>
749
750     </div>
751
752     </div>
753
754     </div>
755
756     </div>
757
758     </div>
759
760     </div>
761
762     </div>
763
764     </div>
765
766     </div>
767
768     </div>
769
770     </div>
771
772     </div>
773
774     </div>
775
776     </div>
777
778     </div>
779
780     </div>
781
782     </div>
783
784     </div>
785
786     </div>
787
788     </div>
789
790     </div>
791
792     </div>
793
794     </div>
795
796     </div>
797
798     </div>
799
800     </div>
801
802     </div>
803
804     </div>
805
806     </div>
807
808     </div>
809
810     </div>
811
812     </div>
813
814     </div>
815
816     </div>
817
818     </div>
819
820     </div>
821
822     </div>
823
824     </div>
825
826     </div>
827
828     </div>
829
830     </div>
831
832     </div>
833
834     </div>
835
836     </div>
837
838     </div>
839
840     </div>
841
842     </div>
843
844     </div>
845
846     </div>
847
848     </div>
849
850     </div>
851
852     </div>
853
854     </div>
855
856     </div>
857
858     </div>
859
860     </div>
861
862     </div>
863
864     </div>
865
866     </div>
867
868     </div>
869
870     </div>
871
872     </div>
873
874     </div>
875
876     </div>
877
878     </div>
879
880     </div>
881
882     </div>
883
884     </div>
885
886     </div>
887
888     </div>
889
890     </div>
891
892     </div>
893
894     </div>
895
896     </div>
897
898     </div>
899
900     </div>
901
902     </div>
903
904     </div>
905
906     </div>
907
908     </div>
909
910     </div>
911
912     </div>
913
914     </div>
915
916     </div>
917
918     </div>
919
920     </div>
921
922     </div>
923
924     </div>
925
926     </div>
927
928     </div>
929
930     </div>
931
932     </div>
933
934     </div>
935
936     </div>
937
938     </div>
939
940     </div>
941
942     </div>
943
944     </div>
945
946     </div>
947
948     </div>
949
950     </div>
951
952     </div>
953
954     </div>
955
956     </div>
957
958     </div>
959
960     </div>
961
962     </div>
963
964     </div>
965
966     </div>
967
968     </div>
969
970     </div>
971
972     </div>
973
974     </div>
975
976     </div>
977
978     </div>
979
980     </div>
981
982     </div>
983
984     </div>
985
986     </div>
987
988     </div>
989
990     </div>
991
992     </div>
993
994     </div>
995
996     </div>
997
998     </div>
999
1000    </div>

```

```

function App() {
  // ...
  </div>

  /* MAXIMUM PRICE */
  <div className="filter-group">
    <label>
      Maximum Price
    </label>
    <input
      type="number"
      value={maxPrice}
      onChange={e => {
        setMaxPrice(
          Number(e.target.value)
        )
      }}
    />
  </div>
</div>

/* -----
PRODUCT COUNT
----- */
<div className="product-count">
  <h2>
    Products Found: {filteredProducts.length}
  </h2>
</div>

/* -----
PRODUCT CARDS
----- */
<div className="products-container">
  {filteredProducts.length === 0 ? (
    <p className="no-products">
      No products found.
    </p>
  ) : (
    filteredProducts.map(product => (
      <div
        className="product-card"
        key={product.id}
      >
        /* PRODUCT IMAGE */
        <img
          src={product.image}
          alt={product.name}
          className="product-image"
        />

        /* PRODUCT NAME */
        <h2>
          {product.name}
        </h2>

        /* CATEGORY */
        <p>
          <strong>
            Category:
          </strong>{ " " }
          {product.category}
        </p>
      </div>
    )
  )}
</div>

```

```

285 function App() {
286     filteredProducts.map(product => (
287
288         /* PRODUCT IMAGE */
289
290         <img
291             src={product.image}
292             alt={product.name}
293             className="product-image"
294         />
295
296         /* PRODUCT NAME */
297
298         <h2>
299             {product.name}
300         </h2>
301
302         /* CATEGORY */
303
304         <p>
305             <strong>
306                 Category:
307             </strong>{" "}
308             {product.category}
309         </p>
310
311         /* PRICE */
312
313         <p>
314             <strong>
315                 Price:
316             </strong>{" "}
317             ₹{product.price}
318         </p>
319
320         /* QUANTITY */
321
322         <p>
323             <strong>
324                 Quantity:
325             </strong>{" "}
326             {product.quantity}
327         </p>
328
329         /* CONDITIONAL RENDERING */
330
331         {product.quantity === 0 ? (
332             <p className="out-of-stock">
333                 ❌ Out of Stock
334             </p>
335         ) : (
336             <p className="in-stock">
337                 ✅ In Stock
338             </p>
339         )}
340     </div>
341 )
342 );
343 </div>
344 );
345 export default App;

```

2.3 App.css

```
App.jsx main.jsx App.css
src > # App.css > header
1  /* GENERAL STYLING */
2
3  {
4      margin: 0;
5      padding: 0;
6      box-sizing: border-box;
7  }
8
9  body {
10     font-family: Arial, Helvetica, sans-serif;
11     background: #eaf3ff;
12     color: #222;
13 }
14
15
16 /* MAIN APP */
17
18 .app {
19     min-height: 100vh;
20     padding-bottom: 50px;
21     background: #eaf3ff;
22     transition: 0.3s;
23 }
24
25
26 /*HEADER*/
27
28 header {
29     background: #d4c99ff;
30     color: white;
31     text-align: center;
32     padding: 40px 20px;
33 }
34
35 header h1 {
36     font-size: 36px;
37     margin-bottom: 10px;
38 }
39
40 header p {
41     font-size: 18px;
42     color: #d8b0afe;
43 }
44
45
46 /* THERE BUTTON */
47
48 .theme-button {
49     margin-top: 20px;
50     padding: 10px 18px;
51
52     border: none;
53     border-radius: 7px;
54
55     background: white;
56     color: #1e3a5f;
57
58     font-weight: bold;
59     font-size: 14px;
60
61     cursor: pointer;
62
63     transition: 0.3s;
64 }
65
66 .theme-button:hover {
67     transform: translateY(-2px);
68 }
69
```

```

71  /* FILTER SECTION*/
72
73  .filters {
74    width: 90%;
75    max-width: 1200px;
76
77    margin: 30px auto;
78    padding: 25px;
79
80    background: #white;
81
82    border-radius: 12px;
83
84    box-shadow:
85      0 4px 12px #rgba(0, 0, 0, 0.1);
86
87    display: grid;
88
89    grid-template-columns:
90      repeat(auto-fit, minmax(220px, 1fr));
91
92    gap: 20px;
93  }
94
95  /*FILTER GROUP */
96
97  .filter-group {
98    display: flex;
99    flex-direction: column;
100  }
101
102  .filter-group label {
103    font-weight: bold;
104    margin-bottom: 8px;
105    color: #333;
106  }
107
108  /* INPUTS AND SELECT */
109
110  .filter-group input,
111  .filter-group select {
112    padding: 12px;
113
114    border: 1px solid #ccc;
115
116    border-radius: 7px;
117
118    font-size: 15px;
119
120    outline: none;
121
122    background: #white;
123    color: #222;
124  }
125
126  .filter-group input:focus,
127  .filter-group select:focus {
128    border-color: #4c99ff;
129
130    box-shadow:
131      0 0 5px #rgba(76, 153, 255, 0.3);
132  }
133
134  /* PRODUCT COUNT*/
135
136  .product-count {
137    width: 90%;
138    max-width: 1200px;
139
140    margin: 20px auto;
141
142    padding: 15px 20px;
143
144    background: #white;
145
146    border-radius: 8px;

```



```

138 # Apprise > header
139 .product-count {
140 }
141
142 .product-count h2 {
143   font-size: 20px;
144   color: #1c3a5f;
145 }
146
147 /* PRODUCTS CONTAINER */
148
149 .products-container {
150   width: 90%;
151   max-width: 1200px;
152   margin: 25px auto;
153   display: grid;
154
155   grid-template-columns:
156     repeat(auto-fit, minmax(250px, 1fr));
157   gap: 25px;
158 }
159
160 /* PRODUCT CARD */
161
162 .product-card {
163   background: #fff;
164   padding: 25px;
165   border-radius: 12px;
166   border: 1px solid #000000;
167   box-shadow:
168     0 4px 10px #000000;
169   transition: 0.3s;
170 }
171
172 .product-card:hover {
173   transform: translateY(-5px);
174   box-shadow:
175     0 8px 18px #000000;
176 }
177
178 /* PRODUCT IMAGE */
179
180 .product-image {
181   width: 100%;
182   height: 180px;
183   object-fit: contain;
184   margin-bottom: 15px;
185   border-radius: 8px;
186 }
187
188 /* PRODUCT NAME */
189
190 .product-card h2 {
191   color: #1c3a5f;
192   margin-bottom: 15px;
193   font-size: 22px;
194 }
195
196 /* PRODUCT DETAILS */
197
198 .product-card p {
199   margin-bottom: 10px;
200   color: #555;
201   line-height: 1.5;
202 }

```

```

src > # App.css > header
230 .product-card p {
231
232     color: #555;
233
234     line-height: 1.5;
235 }
236
237
238
239 /* IN STOCK */
240
241 .in-stock {
242     color: green important;
243
244     font-weight: bold;
245
246     margin-top: 15px;
247 }
248
249
250 /* OUT OF STOCK */
251
252 .out-of-stock {
253     color: red important;
254
255     font-weight: bold;
256
257     margin-top: 15px;
258 }
259
260
261 /* NO PRODUCTS */
262
263 .no-products {
264     grid-column: 1 / -1;
265
266     text-align: center;
267
268     background: white;
269
270     padding: 40px;
271
272     border-radius: 10px;
273
274     color: #777;
275
276     font-size: 18px;
277 }
278
279
280 /* DARK MODE */
281
282 .app.dark {
283     background: #121212;
284     color: #f5f5f5;
285 }
286
287 .app.dark .filters,
288 .app.dark .product-count,
289 .app.dark .product-card,
290 .app.dark .no-products {
291     background: #1e1e1e;
292
293     color: #f5f5f5;
294
295     border-color: #333;
296 }
297
298 .app.dark .filter-group label {
299     color: #ddd;
300 }
301
302 .app.dark .filter-group input,
303 .app.dark .filter-group select {
304     background: #2a2a2a;
305
306     color: white;
307
308     border-color: #555;
309 }
310
311 .app.dark .product-card h2 {
312     color: #6fb1ff;
313 }
314
315 .app.dark .product-card p {

```

```

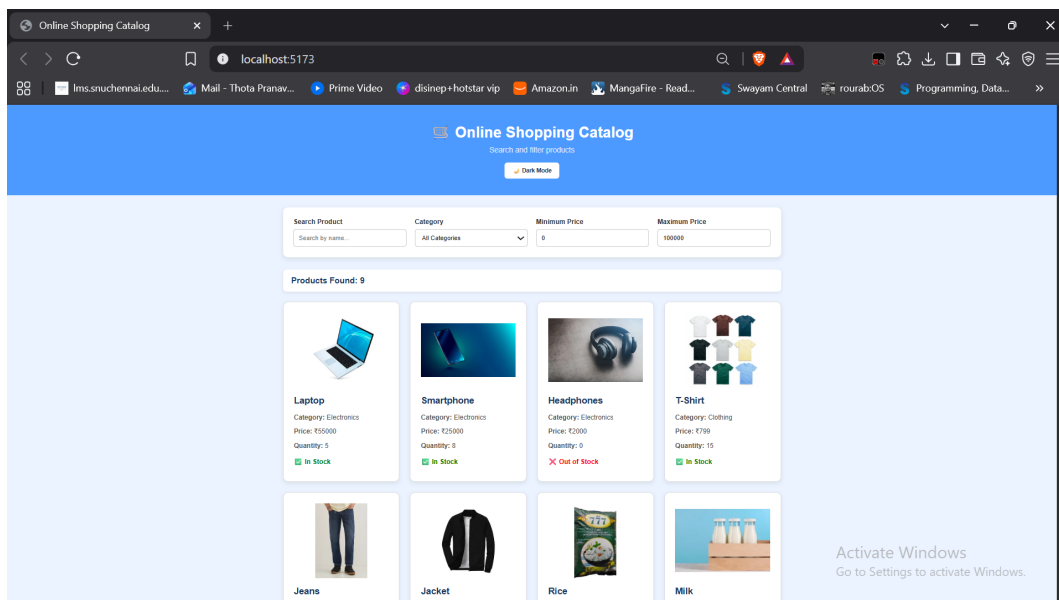
src > # App.css > header
310
311 .app.dark .product-card h2 {
312 |   color: #6fb1ff;
313 }
314
315 .app.dark .product-card p {
316 |   color: #ddd;
317 }
318
319 .app.dark .product-count h2 {
320 |   color: #6fb1ff;
321 }
322
323 .app.dark .no-products {
324 |   color: #ccc;
325 }
326
327 .app.dark .theme-button {
328 |   background: #333;
329 |   color: white;
330 }
331
332
333 /* RESPONSIVE DESIGN */
334
335 @media (max-width: 768px) {
336
337   header h1 {
338 |     font-size: 28px;
339   }
340
341   header p {
342 |     font-size: 16px;
343   }
344
345   .filters {
346 |     width: 95%;
347
348 |     grid-template-columns: 1fr;
349   }
350
351   .product-count {
352 |     width: 95%;
353   }
354
355   .products-container {
356 |     width: 95%;
357
358 |     grid-template-columns: 1fr;
359   }
360
361   .product-image {
362 |     height: 200px;
363   }
364 }

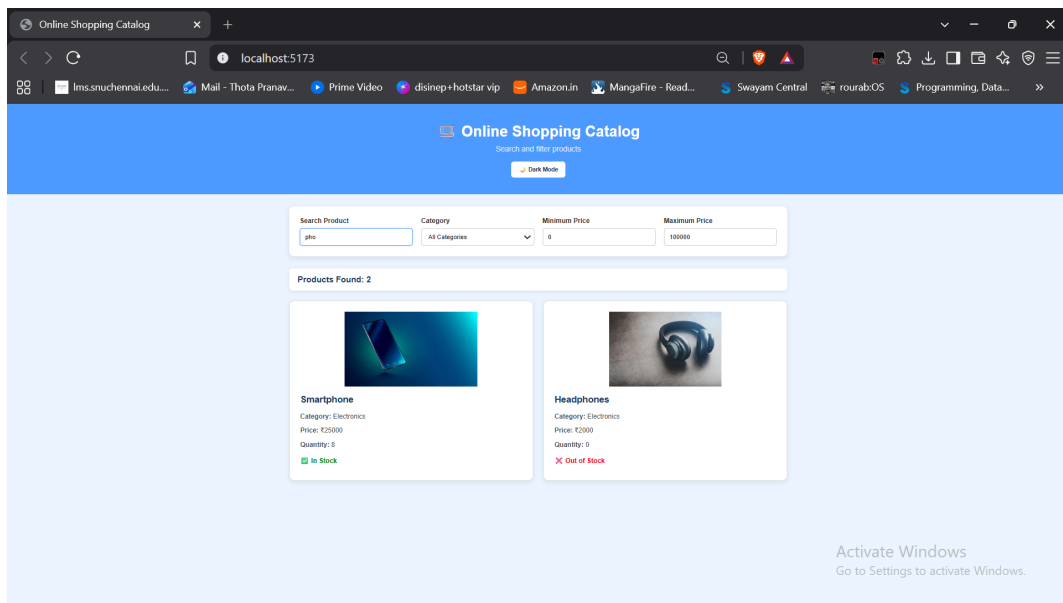
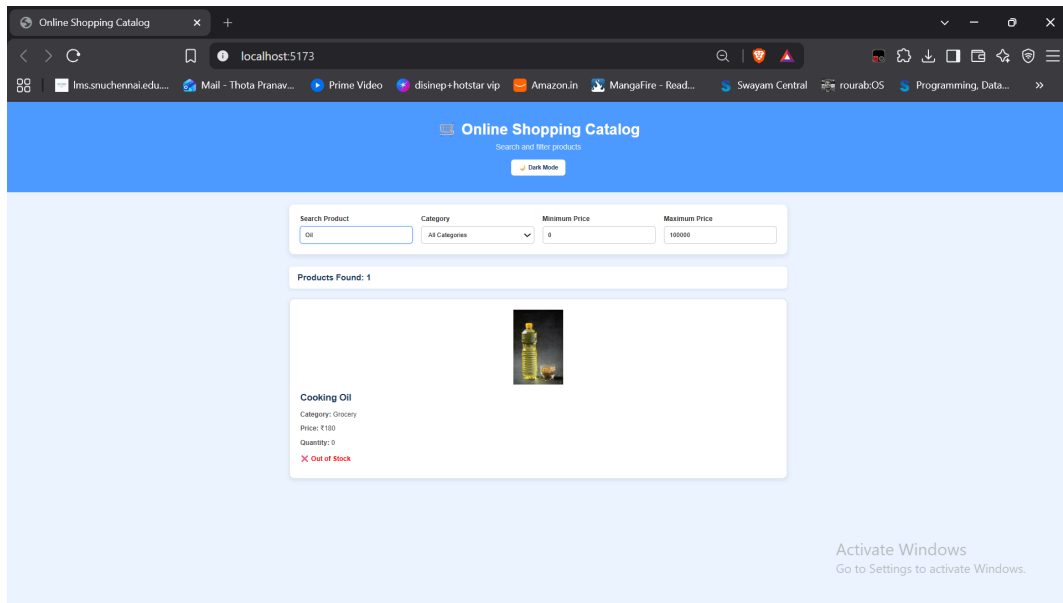
```

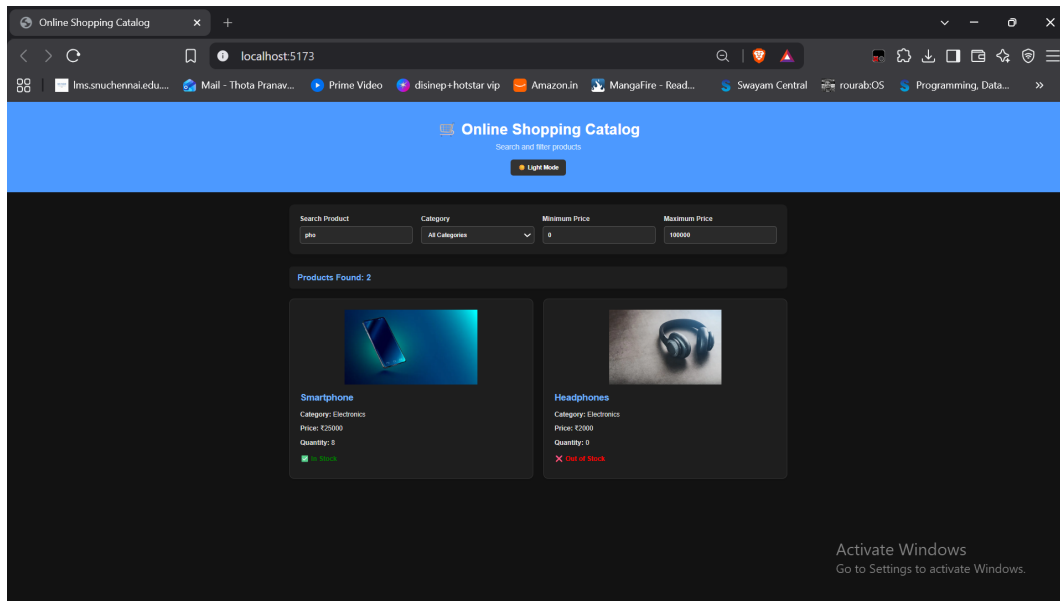
2.4 index.html

```
App.jsx # App.css index.html
index.html > ...
1 <!doctype html>
2 <html lang="en">
3
4 <head>
5   <meta charset="UTF-8">
6
7   <meta name="viewport" content="width=device-width, initial-scale=1.0">
8
9   <title>Online Shopping Catalog</title>
10 </head>
11
12 <body>
13
14   <div id="root"></div>
15
16   <script type="module" src="/src/main.jsx"></script>
17
18 </body>
19
20 </html>
```

3 Webpage







4 Conclusion :

The React-based Online Shopping Product Catalog was successfully developed with dynamic searching, category and price filtering, product count, and conditional stock status. React `useState`, `filter()`, and `map()` were used to create an interactive and responsive product catalog.

Name: Pranav Varshin Thota
Register No.: 24011101116
AIDS-B